

CHAOYOU FU

Assistant Professor

School of Intelligence Science and Technology

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Research Interests

My research interests encompass multimodal intelligence, with particular focus on multimodal large language models. My research primarily involves fundamental multimodal technologies including perception, understanding, interaction, training, and evaluation..

Areas: Multimodal Intelligence

Education and Academic Positions

Nanjing University

Sep.2024 – now

School of Intelligence Science and Technology

– Assistant Professor

Tencent

Jul.2022 – Aug.2024

Youtu Lab

– Senior Researcher

Institute of Automation Chinese Academy of Sciences

Sep.2017 – Jul.2022

National Laboratory of Pattern Recognition

– Ph.D. in Computer Science, Advisor: Ran He

Awards and Honors

IEEE Biometrics Council Best Doctoral Dissertation Award 2023

The World Artificial Intelligence Conference (WAIC) Yunfun Star Award 2025

Chinese Academy of Sciences President's Special Award 2022

CVPR 2023 Outstanding Reviewer (232/7000+) 2023

Outstanding Doctoral Thesis of Chinese Academy of Sciences 2023

Outstanding Doctoral Thesis of Beijing 2023

Publications

1. **Chaoyou Fu**, Xiaoqiang Zhou, Weizan He, et al. “Towards Lightweight Pixel-Wise Hallucination for Heterogeneous Face Recognition.” IEEE Transactions on Pattern Analysis and Machine Intelligence

(**TPAMI**), 2023.

2. **Chaoyou Fu**, Xiang Wu, Yibo Hu, et al. “DVG-Face: Dual Variational Generation for Heterogeneous Face Recognition.” *IEEE Transactions on Pattern Analysis and Machine Intelligence* (**TPAMI**), 2022.
3. **Chaoyou Fu**, Peixian Chen, Yunhang Shen, et al. “MME: A Comprehensive Evaluation Benchmark for Multimodal Large Language Models.” *Advances in Neural Information Processing Systems* (**NeurIPS DB**), 2025.
4. **Chaoyou Fu**, Haojia Lin, Xiong Wang, et al. “VITA-1.5: Towards GPT-4o Level Real-Time Vision and Speech Interaction.” *Advances in Neural Information Processing Systems* (**NeurIPS**), 2025
5. **Chaoyou Fu**, Yibo Hu, Xiang Wu, et al. “High-Fidelity Face Manipulation With Extreme Poses and Expressions.” *IEEE Transactions on Information Forensics and Security* (**TIFS**), 2021.
6. **Chaoyou Fu**, Guoli Wang, Xiang Wu, et al. “Deep Momentum Uncertainty Hashing.” *Pattern Recognition* (**PR**), 2022.
7. **Chaoyou Fu**, Yuhan Dai, Yongdong Luo, et al. “Video-MME: The First-Ever Comprehensive Evaluation Benchmark of Multi-Modal LLMs in Video Analysis.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**), 2025.
8. **Chaoyou Fu**, Yibo Hu, Xiang Wu, et al. “CM-NAS: Cross-Modality Neural Architecture Search for Visible-Infrared Person Re-Identification.” *International Conference on Computer Vision* (**ICCV**), 2021.
9. **Chaoyou Fu**, Xiang Wu, Yibo Hu, et al. “Dual Variational Generation for Low Shot Heterogeneous Face Recognition.” *Neural Information Processing Systems* (**NeurIPS**), 2019.
10. **Chaoyou Fu**, Liangchen Song, Xiang Wu, et al. “Neurons Merging Layer: Towards Progressive Redundancy Reduction for Deep Supervised Hashing”. *International Joint Conference on Artificial Intelligence* (**IJCAI**), 2019.
11. Shukang Yin*, **Chaoyou Fu***, Sirui Zhao, et al. “A survey on multimodal large language models.” *National Science Review* (**NSR**), 2024.
12. Shukang Yin*, **Chaoyou Fu***, Sirui Zhao, et al. “Woodpecker: Hallucination Correction for Multimodal Large Language Models.” *Science China Information Sciences*, 2024.
13. Boyan Duan*, **Chaoyou Fu***, Yi Li, et al. “Cross-Spectral Face Hallucination via Disentangling Independent Factors.” *IEEE/CVF Conference on Computer Vision and Pattern Recognition* (**CVPR**), 2020.
14. Hao Zhu*, **Chaoyou Fu***, Qianyi Wu, et al. “AOT: Appearance Optimal Transport Based Identity Swapping for Forgery Detection.” *Neural Information Processing Systems* (**NeurIPS**), 2020.
15. Zimin Yang, Jian Liang, **Chaoyou Fu**, et al. “Heterogeneous Face Recognition via Face Synthesis With Identity-Attribute Disentanglement.” *IEEE Transactions on Information Forensics and Security* (**TIFS**), 2022.
16. Xiaoqiang Zhou, **Chaoyou Fu**, Huaibo Huang, et al. “Dynamic Graph Memory Bank for Video Inpainting.” *IEEE Transactions on Circuits and Systems for Video Technology* (**TCSVT**), 2024.
17. Linsen Song, Wayne Wu, **Chaoyou Fu**, et al. “Audio-driven Dubbing for User Generated Contents via Style-

aware Semi-parametric Synthesis.” IEEE Transactions on Circuits and Systems for Video Technology (**TCSVT**), 2023.

18. Gege Gao, Huaibo Huang, **Chaoyou Fu**, et al. “Information Bottleneck Disentanglement for Identity Swapping.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
19. Xiong Wang, Yangze Li, **Chaoyou Fu**, et al. “Freeze-Omni: A Smart and Low Latency Speech-to-Speech Dialogue Model with Frozen LLM.” International Conference on Machine Learning (**ICML**), 2025.
20. Yunhang Shen, **Chaoyou Fu**, Peixian Chen, et al. “Aligning and Prompting Everything All at Once for Universal Visual Perception.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2024.
21. Yifan Xu, Mengdan Zhang, **Chaoyou Fu**, et al. “Multi-modal Queried Object Detection in the Wild.” Neural Information Processing Systems (**NeurIPS**), 2023.
22. Linsen Song, Wayne Wu, **Chaoyou Fu**, et al. “Pareidolia Face Reenactment.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2021.
23. Gengyun Jia, Huaibo Huang, **Chaoyou Fu**, et al. “Rethinking Image Cropping: Exploring Diverse Compositions From Global Views.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022.

Academic Services

- **Area Chair: ICLR, ICML**
- **Conference Reviewer: NeurIPS, CVPR, ICCV, ECCV, AAAI, ACM MM, IJCAI**
- **Journal Reviewer: IEEE TPAMI, IJCV, IEEE TIP**